TD04 - Branches + Merge

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Part 1 - Branch isolation

A branch lets me work on a change without touching `main`. Before the merge, the feature branch and `main` are different. After the merge, `main` contains the feature.

Part 2 - Inspect the current branch

```
(base) daphnecortesse@MacBook-Air-de-Daphne esilv-marketpulse-g07-nantais % git status
On branch daphnecortesse
Your branch is ahead of 'origin/daphnecortesse' by 1 commit.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean
(base) daphnecortesse@MacBook-Air-de-Daphne esilv-marketpulse-g07-nantais % git branch
* daphnecortesse
  daphnécortesse
  main
```

I ran `git status` and `git branch`. I was on `daphnecortesse`. The working tree was clean.

The * marks the current branch. The branches are `daphnecortesse`, `daphnécortesse` and `main`.

Part 3 - Create a feature branch

```
(base) daphnecortesse@MacBook-Air-de-Daphne esilv-marketpulse-g07-nantais % git switch main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
(base) daphnecortesse@MacBook-Air-de-Daphne esilv-marketpulse-g07-nantais % git merge daphnecortesse
Updating 5207db8..1e85c67
Fast-forward
 src/main.py | 2 +-
 1 file changed, 1 insertion(+), 1 deletion(-)
(base) daphnecortesse@MacBook-Air-de-Daphne esilv-marketpulse-g07-nantais % git switch -c feature/display-observation-dates
Switched to a new branch 'feature/display-observation-dates'
(base) daphnecortesse@MacBook-Air-de-Daphne esilv-marketpulse-g07-nantais % git branch
  daphnecortesse
  daphnécortesse
* feature/display-observation-dates
  main
```

I switched to `main` with `git switch main`. I merged my TD03 commit with `git merge daphnecortesse`. It was a fast-forward. So `main` has my TD03 work.

Then I created the feature branch with `git switch -c feature/display-observation-dates`. `git branch` shows the * on this new branch. The name describes the work.

Part 4 - Implement the small feature

```
(base) daphnecortesse@MacBook-Air-de-Daphne esilv-marketpulse-g07-nantais % head -3 data/sample/prices.csv
python src/main.py
date,ticker,open,high,low,close,volume
2026-09-01,AAPL,249.20,251.50,248.00,250.00,38000000
2026-09-01,SP500,6592.00,6615.00,6580.00,6600.00,0
=== MarketPulse ===

Instrument
AAPL - Apple Inc.
Observations : 21
First date   : 2026-09-01
Last date    : 2026-09-30
First close  : 250.00 USD
Last close   : 266.20 USD

Benchmark
SP500 - S&P 500
Observations : 21
First date   : 2026-09-01
Last date    : 2026-09-30
First close  : 6600.00
Last close   : 6742.00

Period: 1 month
Interval: Daily
```

The feature shows the first and last observation dates for the instrument and the benchmark. I added three functions in `src/main.py`.

get_first_date: returns the date of the first price.

get_last_date: returns the date of the last price.

print_summary: prints one summary block. I use it for the instrument and for the benchmark.

The CSV has a `date` column, so the functions work. `python src/main.py` runs. AAPL and SP500 both show 21 observations, the first date 2026-09-01 and the last date 2026-09-30.

Part 5 - Inspect and commit the branch change

```
(base) daphnecortesse@MacBook-Air-de-Daphne esilv-marketpulse-g07-nantais % git status
On branch feature/display-observation-dates
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   src/main.py

no changes added to commit (use "git add" and/or "git commit -a")
```

`git status` shows `src/main.py` as modified and not staged. I am on the feature branch.

```
(base) daphnecortesse@MacBook-Air-de-Daphne esilv-marketpulse-g07-nantais % git diff
diff --git a/src/main.py b/src/main.py
index 1fc2e90..5e9b2c2 100644
--- a/src/main.py
+++ b/src/main.py
@@ -26,6 +26,24 @@ def filter_prices(prices, ticker):
     return [row for row in prices if row["ticker"] == ticker]


+def get_first_date(prices):
+    return prices[0]["date"]
+
+
+def get_last_date(prices):
+    return prices[-1]["date"]
+
+
+def print_summary(title, item, prices, currency=""):
+    print(title)
+    print(f"{item['ticker']} - {item['name']}")
+    print(f"Observations : {len(prices)}")
+    print(f"First date   : {get_first_date(prices)}")
+    print(f"Last date    : {get_last_date(prices)}")
+    print(f"First close  : {prices[0]['close']} {currency}".rstrip())
+    print(f"Last close   : {prices[-1]['close']} {currency}".rstrip())
+
+
 def main():
     instruments = load_instruments()
     prices = load_prices()
@@ -36,25 +54,16 @@ def main():
     instrument_prices = filter_prices(prices, instrument["ticker"])
     benchmark_prices = filter_prices(prices, benchmark["ticker"])

-    instrument_latest = instrument_prices[-1]
-    benchmark_latest = benchmark_prices[-1]
-
     print("=== MarketPulse ===")
     print()
-    print("Instrument")
-    print(f"{instrument['ticker']} - {instrument['name']}")
-    print(f"Last price: {instrument_latest['close']} {instrument['currency']}")
+    print_summary(
+        "Instrument", instrument, instrument_prices, instrument["currency"]
+    )
     print()
-    print("Benchmark")
-    print(f"{benchmark['ticker']} - {benchmark['name']}")
-    print(f"Last level: {benchmark_latest['close']}")
+    print_summary("Benchmark", benchmark, benchmark_prices)
     print()
```

`git diff` shows the unstaged changes.

Added: `get_first_date`, `get_last_date` and `print_summary`, and two calls to `print_summary` in `main()`.

Removed: the old print lines, the `instrument_latest` and `benchmark_latest` variables, and the separate "Observations" block. Their information is now in the summary.

I staged the file with `git add src/main.py`. Then I checked `git diff --staged`.

```
(base) daphnecortesse@MacBook-Air-de-Daphne esilv-marketpulse-g07-nantais % git diff --staged
diff --git a/src/main.py b/src/main.py
index 1fc2e90..5e9b2c2 100644
--- a/src/main.py
+++ b/src/main.py
@@ -26,6 +26,24 @@ def filter_prices(prices, ticker):
     return [row for row in prices if row["ticker"] == ticker]


+def get_first_date(prices):
+    return prices[0]["date"]
+
+
+def get_last_date(prices):
+    return prices[-1]["date"]
+
+
+def print_summary(title, item, prices, currency=""):
+    print(title)
+    print(f"{item['ticker']} - {item['name']}")
+    print(f"Observations : {len(prices)}")
+    print(f"First date   : {get_first_date(prices)}")
+    print(f"Last date    : {get_last_date(prices)}")
+    print(f"First close  : {prices[0]['close']} {currency}".rstrip())
+    print(f"Last close   : {prices[-1]['close']} {currency}".rstrip())
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+
 def main():
     instruments = load_instruments()
     prices = load_prices()
@@ -36,25 +54,16 @@ def main():
     instrument_prices = filter_prices(prices, instrument["ticker"])
     benchmark_prices = filter_prices(prices, benchmark["ticker"])

-    instrument_latest = instrument_prices[-1]
-    benchmark_latest = benchmark_prices[-1]
-
     print("=== MarketPulse ===")
     print()
-    print("Instrument")
-    print(f"{instrument['ticker']} - {instrument['name']}")
-    print(f"Last price: {instrument_latest['close']} {instrument['currency']}")
+    print_summary(
+        "Instrument", instrument, instrument_prices, instrument["currency"]
+    )
     print()
-    print("Benchmark")
-    print(f"{benchmark['ticker']} - {benchmark['name']}")
-    print(f"Last level: {benchmark_latest['close']}")
+    print_summary("Benchmark", benchmark, benchmark_prices)
     print()
     print(f"Period: {LOOKBACK_LABEL}")
     print(f"Interval: {INTERVAL_LABEL}")
-    print()
-    print("Observations")
-    print(f"{instrument['ticker']}: {len(instrument_prices)}")
-    print(f"{benchmark['ticker']}: {len(benchmark_prices)}")


 if __name__ == "__main__":
```

It shows the same change. This is what goes into the next commit.

```
(base) daphnecortesse@MacBook-Air-de-Daphne esilv-marketpulse-g07-nantais % git commit -m "feat: display first and last observation dates"
[feature/display-observation-dates 1b7558a] feat: display first and last observation dates
 1 file changed, 22 insertions(+), 13 deletions(-)
```

I committed with a clear message. The commit is 1b7558a. Git reports 1 file changed, 22 insertions and 13 deletions.

```
(base) daphnecortesse@MacBook-Air-de-Daphne esilv-marketpulse-g07-nantais % git status
On branch feature/display-observation-dates
nothing to commit, working tree clean
(base) daphnecortesse@MacBook-Air-de-Daphne esilv-marketpulse-g07-nantais % git log --oneline -5
1b7558a (HEAD -> feature/display-observation-dates) feat: display first and last observation dates
1e85c67 (main, daphnecortesse) feat: add reusable market summaries with first and last closes
5207db8 (origin/main, origin/daphnécortesse, origin/daphnecortesse, origin/HEAD, daphnécortesse) first td session
ba5c9f4 Update Readme
9fd6c8a Update Readme
```

After the commit, the working tree is clean. `git log --oneline -5` shows `1b7558a` at the top, on `feature/display-observation-dates`. `main` is still on `1e85c67`. So the feature commit is not in `main` yet.

Part 6 - Compare feature branch and main

```
(base) daphnecortesse@MacBook-Air-de-Daphne esilv-marketpulse-g07-nantais % git log --oneline -5
1b7558a (HEAD -> feature/display-observation-dates) feat: display first and last observation dates
1e85c67 (main, daphnecortesse) feat: add reusable market summaries with first and last closes
5207db8 (origin/main, origin/daphnécortesse, origin/daphnecortesse, origin/HEAD, daphnécortesse) first td session
ba5c9f4 Update Readme
9fd6c8a Update Readme
(base) daphnecortesse@MacBook-Air-de-Daphne esilv-marketpulse-g07-nantais % git log --oneline --graph --decorate --all
* 1b7558a (HEAD -> feature/display-observation-dates) feat: display first and last observation dates
* 1e85c67 (main, daphnecortesse) feat: add reusable market summaries with first and last closes
* 5207db8 (origin/main, origin/daphnécortesse, origin/daphnecortesse, origin/HEAD, daphnécortesse) first td session
* ba5c9f4 Update Readme
* 9fd6c8a Update Readme
* bfb94a6 init projecttemplate
```

The graph is a straight line. The feature branch is one commit ahead of `main`.

main: at `1e85c67`.

feature/display-observation-dates: at `1b7558a`, the top commit.

```
(base) daphnecortesse@MacBook-Air-de-Daphne esilv-marketpulse-g07-nantais % git diff main..feature/display-observation-dates
diff --git a/src/main.py b/src/main.py
index 1fc2e90..5e9b2c2 100644
--- a/src/main.py
+++ b/src/main.py
@@ -26,6 +26,24 @@ def filter_prices(prices, ticker):
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+def get_first_date(prices):
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+def print_summary(title, item, prices, currency=""):
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     benchmark_prices = filter_prices(prices, benchmark["ticker"])

-    instrument_latest = instrument_prices[-1]
-    benchmark_latest = benchmark_prices[-1]
-
     print("=== MarketPulse ===")
     print()
-    print("Instrument")
-    print(f"{instrument['ticker']} - {instrument['name']}")
-    print(f"Last price: {instrument_latest['close']} {instrument['currency']}")
+    print_summary(
+        "Instrument", instrument, instrument_prices, instrument["currency"]
+    )
     print()
-    print("Benchmark")
-    print(f"{benchmark['ticker']} - {benchmark['name']}")
-    print(f"Last level: {benchmark_latest['close']}")
+    print_summary("Benchmark", benchmark, benchmark_prices)
     print()
     print(f"Period: {LOOKBACK_LABEL}")
     print(f"Interval: {INTERVAL_LABEL}")
-    print()
-    print("Observations")
-    print(f"{instrument['ticker']}: {len(instrument_prices)}")
-    print(f"{benchmark['ticker']}: {len(benchmark_prices)}")


 if __name__ == "__main__":
```

`git diff main..feature/display-observation-dates` shows what the feature branch has and `main` does not.

1. Which change exists on the feature branch? The dates functions and `print_summary`, as shown in the diff.

2. Why is it not yet part of main? The commit is only on the feature branch. `main` still points to the older commit. I have not merged yet.

3. What if the branch were abandoned before merge? `main` would stay unchanged. The feature would never enter `main`.